GitHub and YouTrack Evidence Report

Project Title: Development of an Automated System for Monitoring and Controlling the Implementation of the AITU Development Strategy

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Project Type: Individual Diploma Project

1. Introduction

This document provides evidence of version control, repository organization, and project management activities conducted during the first pre-defense stage of the diploma project.

GitHub was used to organize project deliverables, maintain a history of changes, manage development branches, and document work through pull requests. YouTrack was used to organize project tasks, define expected outcomes and acceptance criteria, assign responsibilities, and track progress.

The evidence presented in this report demonstrates the preparation of the initial project documentation, architecture diagrams, interactive user interface prototype, and verification and evaluation plan.

2. GitHub Repository Structure

The project repository is hosted on GitHub under the name **aitu-strategy-monitoring-system**.

Repository URL: https://github.com/assylayshukirbay/aitu-strategy-monitoring-system

The repository is organized into several directories corresponding to the main deliverables of the first pre-defense:

- **docs/** –Project documentation and first pre-defense report.
- **diagrams/** – System Context, Use Case, and Component diagrams, including editable source files.
- **prototype/** – Interactive HTML prototype and supporting screenshots.
- **tests/** – Verification and evaluation plan, planned test cases, and evaluation templates.
- **README.md** – Project overview, planned functionality, technologies, repository structure, prototype instructions, and project management information.

This structure separates different types of project materials and makes them easier to locate, review, and maintain.

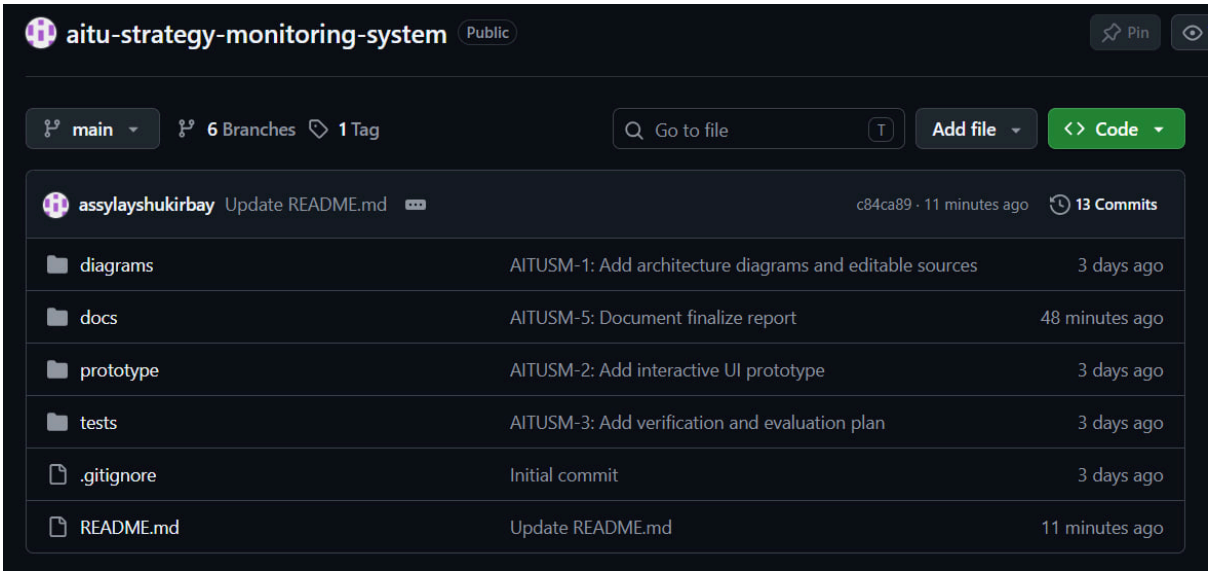


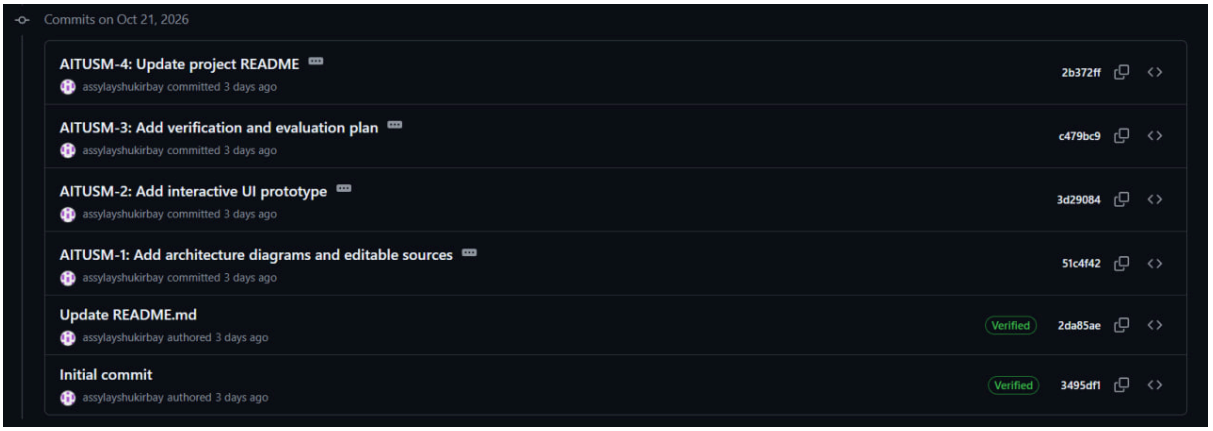
Figure 1. GitHub repository structure showing the main project directories and README.md.

3. Git Commit History

Git was used to track changes to the project files throughout the preparation stage.

Commits document the addition and revision of project materials, including architecture diagrams, the interface prototype, verification documents, the pre-defense report, and repository documentation.

Descriptive commit messages help identify the purpose of each change and establish a traceable development history.



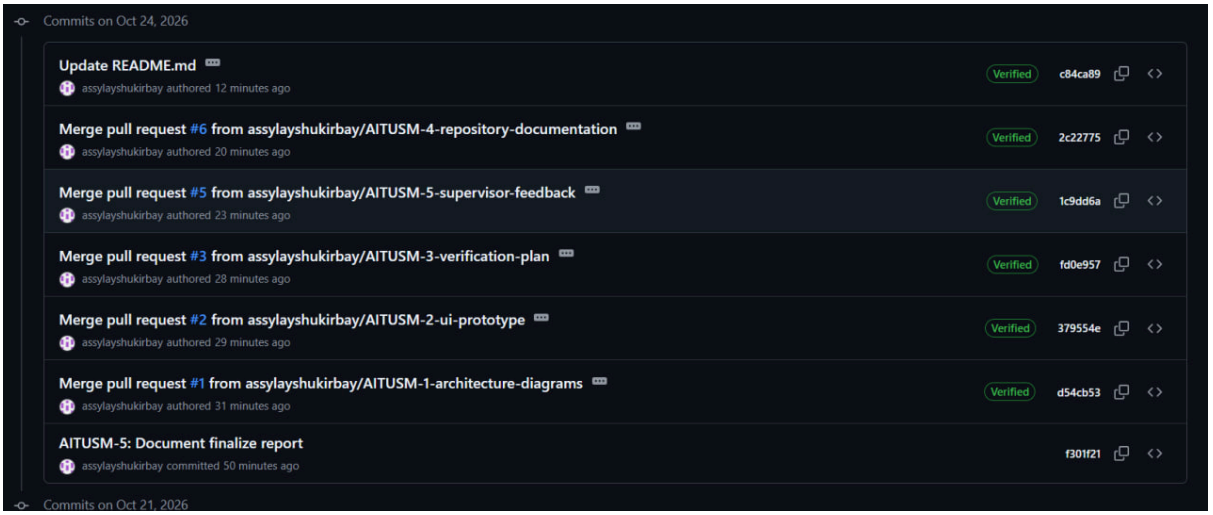


Figure 2. Git commit history showing recorded changes to the project repository.

4. Branch-Based Development

Separate Git branches were used to organize work on different project deliverables.

The development activities were associated with the following YouTrack tasks:

- **AITUSM-1:** Prepare System Architecture Diagrams.
- **AITUSM-2:** Prepare Interactive UI Prototype.
- **AITUSM-3:** Prepare Verification and Evaluation Plan.
- **AITUSM-4:** Prepare GitHub Repository Documentation.
- **AITUSM-5:** Incorporate Supervisor Feedback into the Pre-Defense Report.

This approach allowed project changes to be prepared and reviewed separately before integration into the main repository.

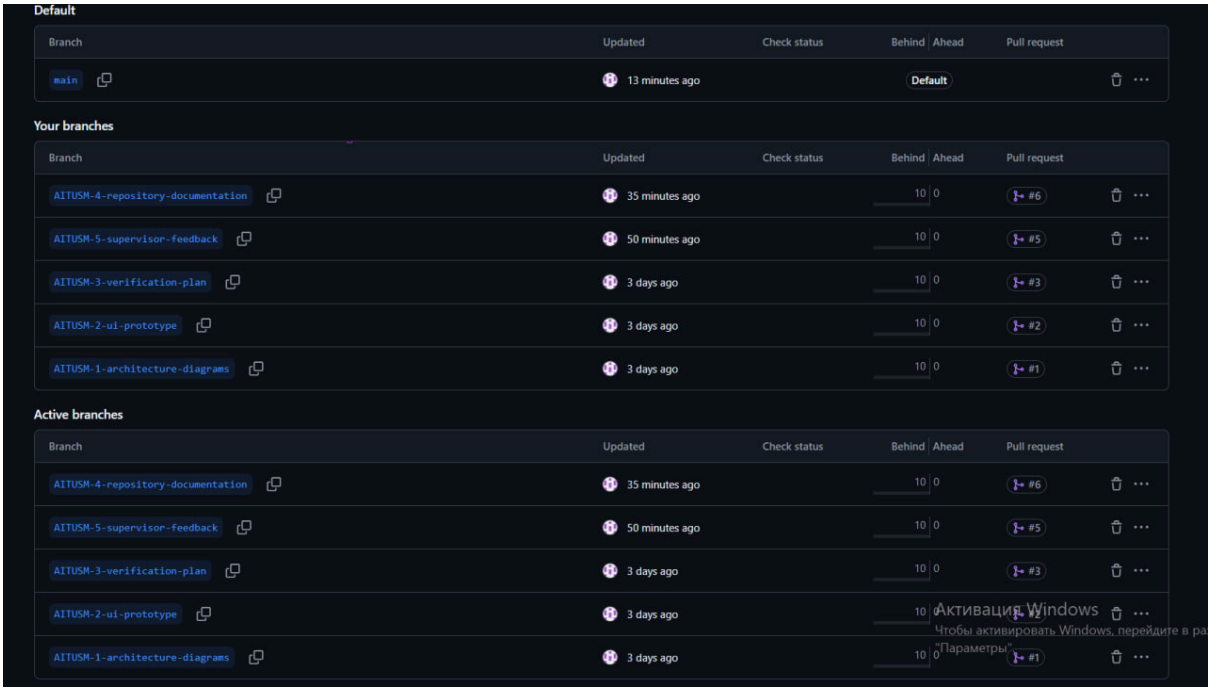


Figure 3. GitHub branches used to organize individual project deliverables.

5. Pull Requests and Integration

Pull requests were used to document and integrate changes into the main branch.

Each pull request provides a record of the proposed changes, associated files, commit history, and integration status.

The project includes pull requests for architecture diagrams, the interactive prototype, verification materials, the revised report, and repository documentation.

The repository documentation task is represented by Pull Request #6, which replaced an earlier closed Pull Request #4.

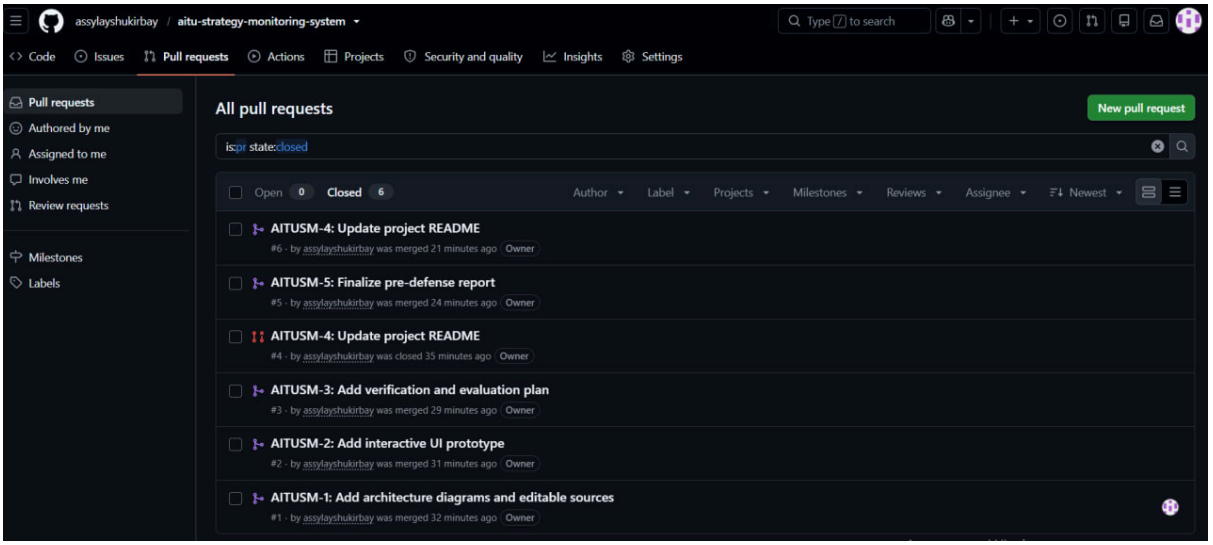


Figure 4. Pull request history demonstrating the integration of project deliverables into the main branch.

6. First Pre-Defense Git Milestone

A Git milestone was prepared using the tag **pre-defense-1**, targeting the main branch.

This milestone identifies the repository version associated with the first pre-defense and provides a reference point for subsequent development stages.

The milestone includes the prepared project documentation, architecture diagrams, interactive HTML prototype, verification and evaluation materials, and updated README.

The backend, database integration, and full system implementation remain planned for later stages.

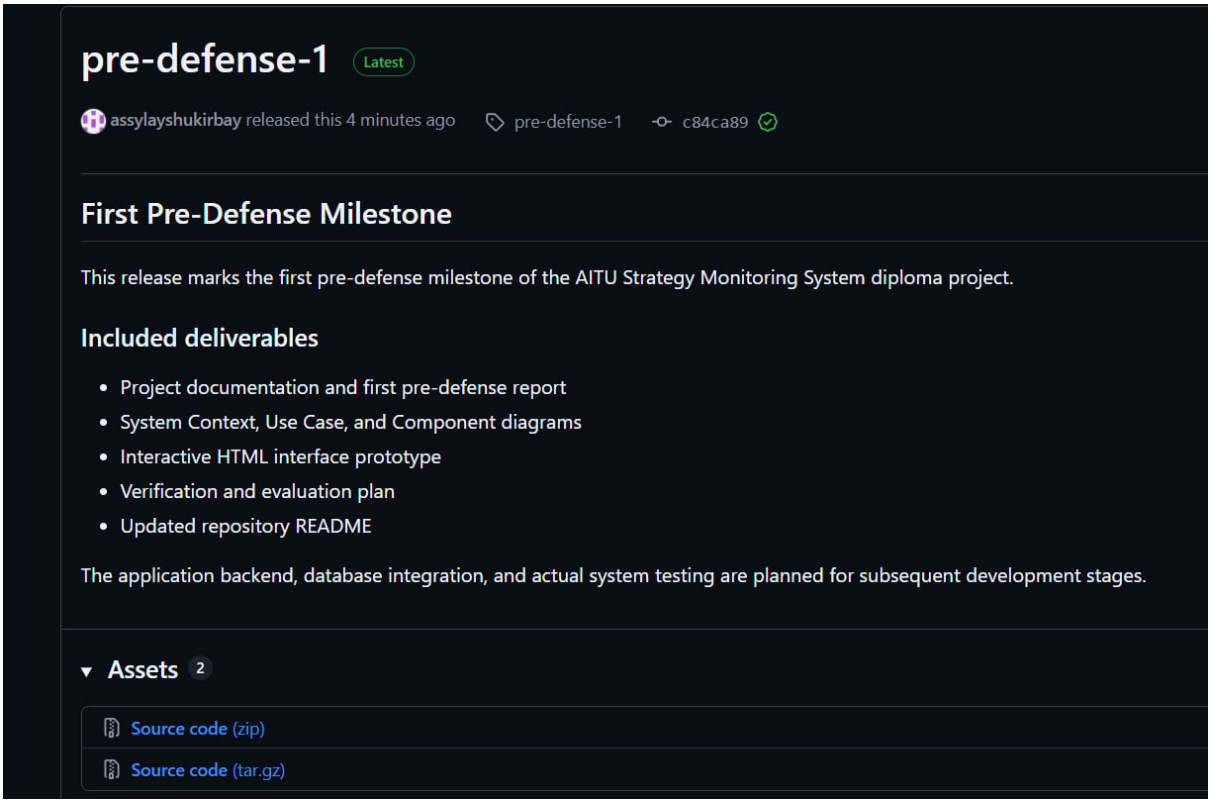


Figure 5. First pre-defense milestone identified by the Git tag pre-defense-1.

7. YouTrack Project Management

YouTrack was used to organize and monitor the project tasks.

YouTrack Project URL: https://assylay.youtrack.cloud/projects/AITUSM

The project uses the identifier **AITUSM**. Six tasks were created to cover the preparation and documentation activities required for the first pre-defense.

The task management system records task titles, assigned responsibilities, priorities, statuses, and supporting information.

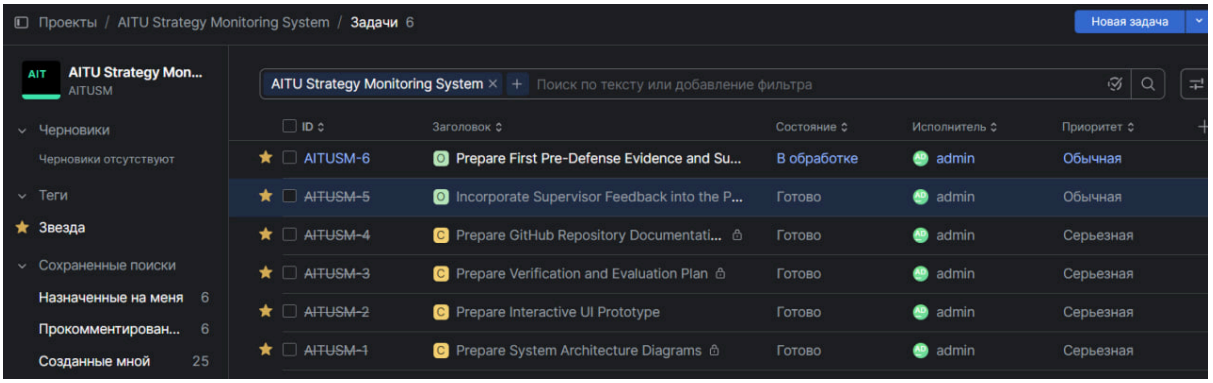


Figure 6. YouTrack task overview showing the project work items, assignee, priorities, and progress statuses.

8. Task Requirements and Acceptance Criteria

Project tasks were documented with descriptions, expected outcomes, and acceptance criteria.

For example, **AITUSM-1 – Prepare System Architecture Diagrams** covers the preparation of the System Context, Use Case, and Component diagrams and their editable source files.

Defining acceptance criteria provides a clear basis for checking whether the required deliverables have been prepared.

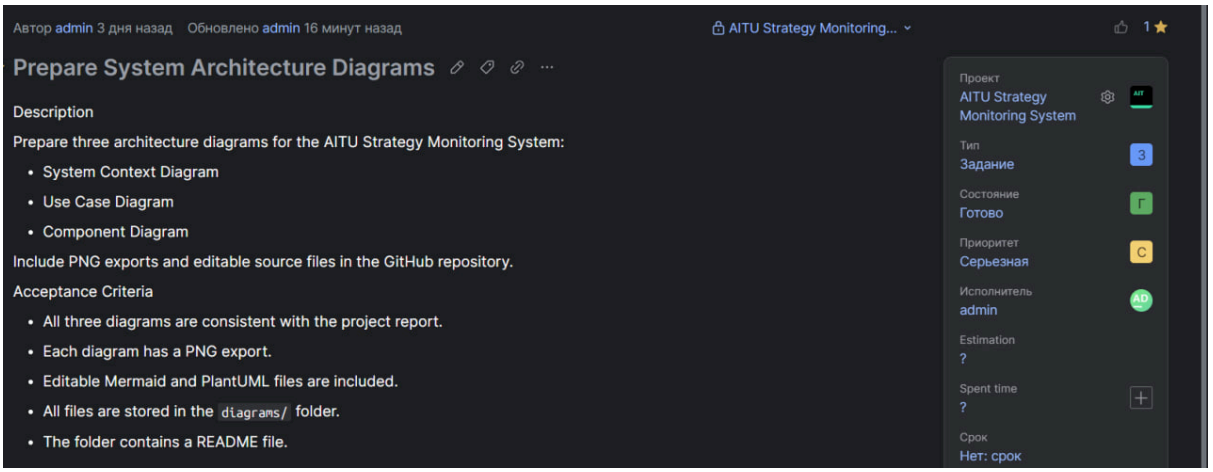


Figure 7. Example of a YouTrack task with defined deliverables and acceptance criteria.

9. Traceability Between YouTrack and GitHub

The relationship between project management and version control is documented through task identifiers, commit messages, and pull request links.

For example, task **AITUSM-1** corresponds to GitHub Pull Request #1, which documents the architecture diagram deliverables.

This traceability allows a reviewer to connect an assigned task with the corresponding repository changes and supporting evidence.



Figure 8. Evidence of traceability between a YouTrack task and its corresponding GitHub pull request.

10. Supervisor Feedback and Report Revisions

Supervisor feedback was incorporated into the first pre-defense report and documented under **AITUSM-5 – Incorporate Supervisor Feedback into the Pre-Defense Report**.

Written feedback received on October 24, 2026 clarified that the existing strategy monitoring process at AITU relies on manual information collection, calculation of implementation progress, and report preparation.

Based on this feedback, Section 2 of the report was revised to explain the current process and the relevance of the proposed automated system. Section 11 was updated to document the supervisor's comments, completed revisions, and decisions remaining for subsequent stages.

The supervisor indicated that the prepared report was sufficient for the first pre-defense, subject to meeting the assignment requirements. The final selection of implementation technologies and reporting tools remains open for further discussion.

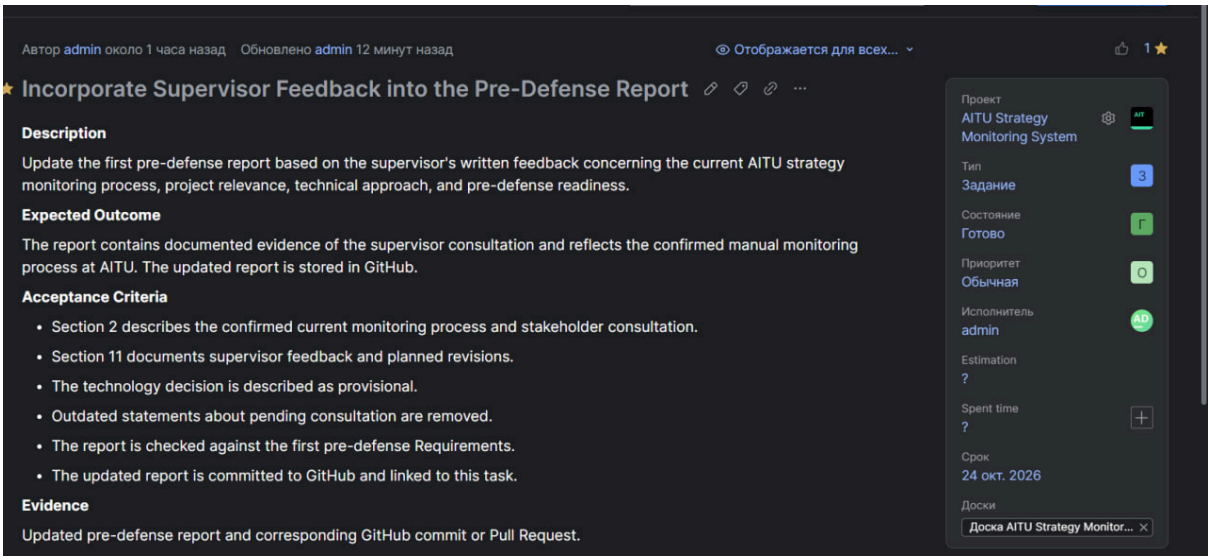


Figure 9. YouTrack documentation of supervisor feedback and corresponding report revisions.

11. Final Evidence Preparation Task

Task **AITUSM-6 – Prepare First Pre-Defense Evidence and Submission Materials** was created to organize the final submission evidence.

The task covers verification of repository deliverables, creation of the first pre-defense Git milestone, collection of GitHub and YouTrack screenshots, and preparation of the supporting evidence document.

This task provides a separate record of the final documentation and submission preparation activities.

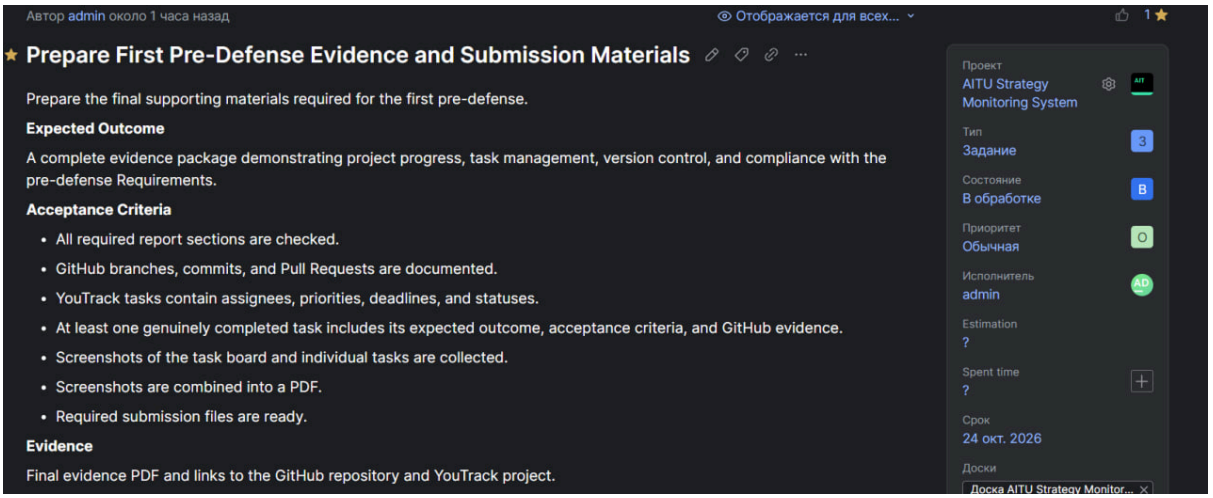


Figure 10. YouTrack task created to manage preparation of the first pre-defense evidence package.

12. Conclusion

The first pre-defense stage established an organized foundation for the AITU Strategy Monitoring System diploma project.

GitHub provides a structured repository, documented changes, separate development branches, pull request history, and a milestone reference. YouTrack supports task planning, progress tracking, responsibility assignment, and traceability between project activities and repository deliverables.

The current project materials include the initial documentation, architecture diagrams, interactive interface prototype, and verification and evaluation plan.

These artifacts support the first pre-defense and provide a starting point for subsequent implementation, integration, testing, and evaluation activities.